
OpenWAM

Developer Guide

A source map: each component and behaviour, and the file that owns it

The reference companion to the *OpenWAM User Guide* and *Glossary*

Contents

Purpose and conventions	1
1 Build and run	1
2 Solver core (1-D interior and boundaries)	1
3 Ducts, junctions, ends (Nodes tab)	2
4 Connections tab	2
5 0-D volumes (0D Elements tab)	3
6 Engine (Engines tab)	3
7 Turbomachinery (Turbocharger tab)	3
8 Control (Control tab)	3
9 Output and results	4
10 External / co-simulation and miscellany	4
11 Where behaviour lives, by question	4

Purpose and conventions

This guide answers one question: a developer wants the file and function that owns a component or a behaviour. It is a lookup index, not a narrative — there is no change history and no bug commentary here (use `git log` and `git blame` for those). All paths are under `Source/`.

Two conventions hold across the component classes. `LeeDatosIniciales` reads that class's fields from the `.WAM`; `CalculaCondicionContorno` (or the class's per-step method) is its physics.

Anchor on function names, not line numbers — line numbers drift, so `grep` the name. For the `.WAM` line layout of any component, and the node and deposit type codes, see `Notes/WAM-File-Format-Guide.md`. for a broader tour of the code, see `Notes/OpenWAM_Codebase_Guide.md`.

1 Build and run

- Entry point: `OpenWAM.cpp::main` → `TOpenWAM::ReadInputData(argv[1])` (reads the whole `.WAM`) → the time loop.
- Orchestration and read order: `TOpenWAM.cpp`. Batch and time control: `TTimeControl.cpp`, `TCGestorWAM.cpp`.
- Build: `CMake + Ninja + MSVC (x86)`. The parallel target `build-parallel` uses `OpenMP` and is bit-identical to the serial build.

2 Solver core (1-D interior and boundaries)

- **1-D duct solver** (RoeM flux, MUSCL, TVD-RK3; also Lax–Wendroff+FCT; area source; friction/heat): `1DPipes/TTubo.cpp`. The key entry is the flux/residual/step methods and the `CalculaVariablesFundamentales` dispatch.

- **Boundary-condition base:** Boundaries/TCondicionContorno.cpp; the method-of- characteristics transforms live in TTubo.
- **Junctions (GJM ghost cell):** Boundaries/TCCRamificacion.cpp::CalculaCondicionContorno.
- **Gas properties / mixture γ , R, Cv; global constants and clamps:** Includes/Globales.h, Includes/Constantes.h (namespace __cons).
- **Per-step orchestration** (global dt $\rightarrow$ interior $\rightarrow$ boundaries $\rightarrow$ update $\rightarrow$ gas properties): the Common path in TOpenWAM.cpp; the Independent path is in the same file.

3 Ducts, junctions, ends (Nodes tab)

UI element / behaviour	Source
Duct (pipe)	1DPipes/TTubo.cpp
Ramification (Y junction)	Boundaries/TCCRamificacion.cpp
Union of two pipes	Boundaries/TCCUnionEntreTubos.cpp
Pressure loss (laminar / turbulent)	Boundaries/TCCPerdidadePresion.cpp (physics in BoundaryFunctions.h, stPerdPresAd/L)
Open end (Extreme to Atmosphere)	Boundaries/TCCDescargaExtremoAbierto.cpp
Anechoic end	Boundaries/TCCExtremoAnecoico.cpp
Closed end	Boundaries/TCCExtremoCerrado.cpp
Prescribed static-P / const-P&T inlet	Boundaries/TCCPreVble.cpp
Pressure pulse inlet	Boundaries/TCCPulso.cpp, Boundaries/TEntradaPulso.cpp
Injection extreme	Boundaries/TCCExtremoInyeccion.cpp
Screw compressor (supercharger BC)	Boundaries/TCCCompresorVolumetrico.cpp + BoundaryFunctions.h, stComprVol
External / CFD / MATLAB coupling	Boundaries/TCCExternalConnection*.cpp, TCFDConnection.cpp, Extern/TMatlab.cpp

4 Connections tab

UI element / behaviour	Source
Valve (poppet, 4T)	Connections/TValvula4T.cpp (base TTipoValvula.cpp)
Controlled valve / external	Connections/TValvulaContr.cpp, Connections/TCDExterno.cpp
Throttle (butterfly)	Connections/TMariposa.cpp
Port (2T)	Connections/TLumbrera.cpp
Rotary disk (2T intake)	Connections/TDiscoRotativo.cpp
Reed valve	Connections/TLamina.cpp
Wastegate (mechanical)	Connections/TWasteGate.cpp
Turbine stator / rotor	Connections/TEstatorTurbina.cpp, Connections/TRotorTurbina.cpp
Node $\leftrightarrow$ deposit	Boundaries/TCCUnionEntreDepositos.cpp, Boundaries/TCCDeposito.cpp
Discharge-coefficient helper	Connections/TCDFijo.cpp

5 0-D volumes (0D Elements tab)

UI element / behaviour	Source
Deposit base	ODModels/TDeposito.cpp
Constant-volume chamber	ODModels/TDepVolCte.cpp (+ TDepVolCteBase.cpp)
Variable-volume chamber	ODModels/TDepVolVariable.cpp
Venturi	ODModels/TVenturi.cpp
Directional junction	ODModels/TUnionDireccional.cpp

6 Engine (Engines tab)

UI element / behaviour	Source
Engine block (4T/2T)	Engine/TBloqueMotor.cpp
Cylinder base (heat release, Woschni, IMEP)	Engine/TCilindro.cpp
4-stroke cylinder	Engine/TCilindro4T.cpp
2-stroke cylinder (scavenging)	Engine/TCilindro2T.cpp
Cylinder BC (valve ↔ cylinder)	Boundaries/TCCcilindro.cpp
Heat-release law (FQL / Wiebe)	Extern/Tfql.cpp
ACT predictive diesel + NOx	Act/ACT_Sub_DLL.cpp, Act/INJECTION_RATE.cpp
EGR coefficients	Extern/TEGRV.cpp

7 Turbomachinery (Turbocharger tab)

UI element / behaviour	Source
Turbo shaft	Turbocompressor/TEjeTurbogruppo.cpp
Compressor + map	Turbocompressor/TCompresor.cpp, BC Boundaries/TCCCompresor.cpp, inlet TCCEntradaCompresor.cpp
Turbine (0-D + map)	ODModels/TTurbina.cpp, TTurbinaSimple.cpp, TTurbineMap.cpp
VGT position / iso-speed	ODModels/TTurbPosition.cpp, TISOspeedLine.cpp; control Extern/TTGV.cpp, TControlK.cpp
Twin-flow turbine	ODModels/TTurbinaTwin.cpp
Acoustic turbine	ODModels/TAcousticTurbine.cpp
Turbo-pipe coupling	Turbocompressor/TCompTubos.cpp, TCompTubDep.cpp

8 Control (Control tab)

UI element / behaviour	Source
Controller base	Control/TController.cpp
PID	Control/TPIDController.cpp

UI element / behaviour	Source
Table-1D / interpolation	Control/TTable1D.cpp (+ TTable.cpp); file-read format in LeeDatosIniciales
Sensor	Control/TSensor.cpp
Switch / decisor	Control/TDecisor.cpp
Gain	Control/TGain.cpp

9 Output and results

All results — the AVG/INS blocks, space-time maps, and the per-element case-id selection — live in `Output/TOutputResults.cpp`. The space-time block reader and writers are `ReadSpaceTimeResults`, `HeaderSpaceTimeResults`, `PrintSpaceTimeResults`, and `WriteSpaceTime`.

10 External / co-simulation and miscellany

- MATLAB / co-simulation, engine-speed control, fuel and injection control, spring, transfer-coefficient fitting: `Extern/` — `TMatlab.cpp`, `TRegimenMotor.cpp`, `TControlFuel.cpp`, `TControlInyeccion.cpp`, `TMuelle.cpp`, `TAjusteTransCalorCil.cpp`, `TCoefDescarga.cpp`.
- **The MATLAB tie-in is disabled** (non-functional to the user). The live MATLAB-Engine ECU bridge (`TMatlab`) is not instantiated: `ECU = new TMatlab(...)` is commented out in `TCalculoExtern.cpp`, no MATLAB library is linked, and the code targets an obsolete Engine API. Reviving it means uncommenting, porting to the modern API, and linking the library.
- Diesel particulate filter: `DPF/`. Concentric pipe: `Concentric Pipe/`. Math helpers: `Math_-wam/`.

11 Where behaviour lives, by question

UI element / behaviour	Source
How is a field read from the WAM?	that class's <code>LeeDatosIniciales</code>
What is the WAM line layout?	<code>Notes/WAM-File-Format-Guide.md</code> (topology + node/deposit codes)
Which numerical method / positivity / Courant?	<code>1DPipes/TTubo.cpp</code>
Junction energy behaviour?	<code>Boundaries/TCCRamificacion.cpp</code>
Read order of the whole file / how components link?	<code>TOpenWAM::ReadInputData</code> and the BC-list build loops in <code>TOpenWAM.cpp</code>